



1. The smallest integer than can be represented by an 8-bit number in 2's complement form is

(a) -256 (b) -128  
(c) -127 (d) 0

**Computer – Number System (2's Complement) – 2021**

2. 2's complement of 1011011 is

(a) 0100011 (b) 0110101  
(c) 0100111 (d) 0100101

**Computer – 2's Complement – 2021**

3. The decimal equivalent of the binary number  $(1011.01)_2$

(a)  $(11.375)_{10}$  (b)  $(10.123)_{10}$   
(c)  $(11.25)_{10}$  (d)  $(9.23)_{10}$

**Computer – Binary to Decimal Conversion – 2021**

4. The product of two four bit positive binary numbers 1011 and 0011 is \_\_\_\_\_

(a) 1000011 (b) 100001  
(c) 100101 (d) 100111

**COMPUTER – Binary Multiplication – 2022**

5. Representation of -11 in sign – and magnitude is

(a) 11011 (b) 010111  
(c) 10011 (d) 01110

**COMPUTER – Number Representation – 2022**

6. 1001 is 2's complement representation of \_\_\_\_\_.

(a) -7 (b) +9  
(c) +6 (d) -6

**COMPUTER – 2's Complement – 2022**

7. Given below are two statements:

**Statement I:** 011010 is 2's complement representation of -37.

**Statement II:**  $111_{(8)}$  is octal representation of 73.

In the light of the above statements, choose the most appropriate answer from the options given below:

(a) Both Statement I and Statement II are true  
(b) Both Statement I and Statement II are false  
(c) Statement I is true but Statement II is false  
(d) Statement I is false Statement II are true

**COMPUTER – Number System – 2022**

8. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

**Assertion A:** Binary information is represented in digital in terms of 'zeros' and 'ones'.

**Reason R:** Digital computers are electronic devices, which operate using electrical voltages.

In the light of the above statements, choose the most appropriate answer from the options given below:

(a) Both A and R are correct and R is the correct explanation of A

(b) Both A and R are correct and R is not the correct explanation of A

(c) A is correct but R is not correct.

(d) A is correct but R is correct.

**COMPUTER – Computer Fundamentals – 2022**

9. A 2's-complement adder- subtractor can add or subtract binary numbers. Sign-magnitude numbers represent \_\_\_\_\_ decimal numbers, and 2's complements stand for \_\_\_\_\_ decimal numbers.

(a) hexa, sign (b) sign, hexa  
(c) positive, negative (d) negative, positive

**COMPUTER – Number Representation – 2023**

- 10.

A. If  $(12P)_3 - (123)_p$ , then value of P is infeasible.

B. The simplified sum of product form of the Boolean expression is  $(P + \bar{Q} + \bar{R}). (P + \bar{Q} + R). (P + Q + \bar{R})$  is  $(P + \bar{Q}. \bar{R})$ .

C. The minimum number of D flip-flops needed to design a mod (258) counter is 8.

Choose the correct answer from the options given below:

(a) A only (b) A and B only  
(c) A and C only (d) C only

**COMPUTER – Digital Electronics – 2023**

11. The number of 1's in the binary representation of  $(3 \times 4096 + 15 \times 256 + 5 \times 16 + 3)$  are

(a) 8 (b) 29 (c) 10 (d) 12

**COMPUTER – Binary Numbers – 2023**

12. Given the following binary number in 32-bit (single precision) IEEE-754 format:

00111110011011010000000000000000

The decimal value closest to this floating-point number is

(a)  $1.45 \times 10^1$  (b)  $1.45 \times 10^{-1}$   
(c)  $2.27 \times 10^{-1}$  (d)  $2.27 \times 10^1$

**COMPUTER – IEEE-754 – 2023**

13. The range of integers that can be represented by a 2's complement number system is \_\_\_\_\_. Where n is number of bits in number

(a)  $-2^{n-1}$  to  $2^{n-1} - 1$  (b)  $-(2^{n-1} - 1)$  to  $(2^{n-1} - 1)$   
(c)  $-2^{n-1}$  to  $2^{n-1}$  (d)  $-(2^{n-1} + 1)$  to  $(2^{n-1} - 1)$

**Computer Organization – Number System  
(2's Complement Range) – 2025**

14. Given  $\sqrt{(224)_r} = (13)_r$  where r is the radix. The value of r is \_\_\_\_\_

(a) 10 (b) 8 (c) 5 (d) 6

**Number System – Radix/Base System – 2025**





15. Which of the following statement is true about IEEE754 representation of +0 and -0?

**(CUET-PG 2025) Floating Point (IEEE-754) – 2025**

- (a) have the same sign bit, but different mantissa bits.  
(b) differ only in the sign bit; their exponent and mantissa bits are identical.  
(c) have the same representation because they are mathematically equivalent.  
(d) differ in both their exponent and mantissa bits.
16. Perform the arithmetic addition of the two decimal numbers given in **List-I** using the signed-complement system. Match the corresponding output of **List-I** with binary number representation given in **List-II**.

| List-I      | List-II         |
|-------------|-----------------|
| (A) +6, +13 | (I). 00000111   |
| (B) -6, +13 | (II). 00010011  |
| (C) +6, -13 | (III). 11101101 |
| (D) -6, -13 | (IV). 1111101   |

**(CUET-PG 2025) Signed Complement – 2025**

Choose the **correct** answer from the options given below:

- (a) (A) - (II), (B) - (I), (C) - (III), (D) - (IV)  
(b) (A) - (II), (B) - (I), (C) - (IV), (D) - (III)  
(c) (A) - (I), (B) - (II), (C) - (IV), (D) - (III)  
(d) (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

## ANSWER KEY

| 1.       | 2.       | 3.       | 4.       | 5.       | 6.       | 7.       | 8.       | 9.       | 10.      |
|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| <b>b</b> | <b>d</b> | <b>c</b> | <b>b</b> | <b>a</b> | <b>a</b> | <b>d</b> | <b>a</b> | <b>c</b> | <b>b</b> |
| 11.      | 12.      | 13.      | 14.      | 15.      | 16.      |          |          |          |          |
| <b>d</b> | <b>c</b> | <b>a</b> | <b>c</b> | <b>b</b> | <b>b</b> |          |          |          |          |

